

Usability Evaluation of E-Learning Website

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Abstract— E-learning programs are designed to provide knowledge effectively, share information, and support learners in their educational endeavors. The success of these programs relies heavily on the usability of the e-learning applications, which must adhere to core usability principles. This study, in connection with the evaluation of the usability of the MOODLE e-learning platform, focuses specifically on users with various disabilities, the demographic that has been insufficiently addressed in prior research, aimed at the above facts. The research was conducted in two phases: the first involved reviewing literature and expert evaluations to establish and refine usability criteria. These criteria were then used in the second phase to develop a conceptual model for assessing usability based on effectiveness, efficiency, and satisfaction. Effectiveness and efficiency were evaluated by using the think-aloud protocol, while satisfaction was measured through a questionnaire. A total of 100 respondents participated in the study, and the collected data were analyzed by using Partial Least Squares Structural Equation Modeling (PLS-SEM). This research not only fills a critical gap in existing usability studies but also aims to enhance the inclusivity and accessibility of the MOODLE platform for all users, ensuring that e-learning solutions are equitable and effective for a diverse range of learners.

Keywords— E-learning, MOODLE, Smart PLS, Think-aloud protocol, WCAG

I. INTRODUCTION

The history of learning and e-learning is rich and diverse, spanning centuries of human development. Traditional learning methods have evolved, from oral traditions and apprenticeships to formalized education systems. However, the emergence of digital technology has revolutionized the way we learn, giving rise to e-learning.

The Industrial Revolution heralded a transformative era in education, characterized by the establishment of public schools and the standardization of curricula, marking a shift towards mass education and innovative teaching methodologies [1]. The advent of computers in the 20th century further revolutionized learning, giving rise to e-learning platforms like the PLATO (Programmed Logic for Automated Teaching Operations) system, pioneered at the University of Illinois in the 1960s [2]. This groundbreaking system facilitated student interaction with computer-based lessons and assessments, laying the foundation for the proliferation of modern e-learning platforms.

In recent years, e-learning has emerged as a pivotal tool for educational institutions worldwide, allowing for flexible, accessible, and diverse learning experiences. Universities in Sri Lanka have increasingly adopted online platforms to

conduct lessons, offering students the convenience of accessing education from any location. However, the effectiveness of these e-learning systems is contingent on their usability, particularly for users with various disabilities. These disabilities may include visual impairments, hearing impairments, motor disabilities, and cognitive impairments, each presenting unique challenges in navigating and utilizing digital learning environments. Evaluating the performance of an e-learning system is thus crucial to ensure that it meets the diverse needs of its users.

The primary objective of this study is to assess the usability of e-learning websites, with a specific focus on the MOODLE platform. The choice to evaluate MOODLE over other platforms is rooted in its prominent use across educational institutions worldwide, as well as its open-source design, which allows for flexibility and innovation in usability studies. MOODLE offers a range of pedagogical services, including online lectures, tutorials, exams, e-library resources, and facilities for downloading recorded lecture materials, all aimed at enhancing user engagement and learning experiences. Despite its widespread use, university websites often face significant usability issues, affecting students, lecturers, and other visitors.

Common usability problems on e-learning websites include complex navigation structures, ambiguous labels, and color contrast issues, which hinder the user experience and access to vital information. Inconsistent design elements across pages contribute to a disjointed interface, diminishing the platform's professional appeal. Furthermore, the lack of mobile optimization results in slow loading times and broken layouts on smartphones and tablets. Additionally, non-compliance with web accessibility standards limits access for individuals with disabilities [3].

This research employs the ISO 9241-11 framework, which defines usability through three key factors: effectiveness, efficiency, and satisfaction [4]. By evaluating these aspects, the study aims to provide insights into how well the MOODLE platform supports the learning process for users with disabilities and identify areas for improvement to enhance their overall learning experience.

II. LITERATURE REVIEW

A. Usability Attributes and Model

In today's digital era, learning websites have become crucial platforms for education, necessitating the optimization of their usability as online learning demand increases. Evaluating the usability of these sites is essential for enhancing effectiveness and user satisfaction. The concept of usability has evolved, with various terms like Human-

Computer Interaction (HCI) and User Experience (UX) emerging, but a universally accepted definition remains elusive [4] [5]. Initially, usability attributes were categorized into frameworks like McCall's and Gould's, but subsequent research identified additional dimensions such as effectiveness, user satisfaction, accessibility, and learnability. Broader perspectives on usability emerged, considering technical, cognitive, social, and design aspects [4]. Guidelines like WCAG and standards such as ISO 9241-11 have further advanced website usability evaluation, shaping the discipline significantly [4].

Model	Variables					
	Ease of use	Clarity	Accessibility	Attractiveness	Flexibility	Security
Bass et al (2003)	✓	✓				✓
Saffah, Donyae, Padda & Kline (2006)	✓		✓	✓	✓	✓
Hix & Hartson (1993)	✓	✓		✓	✓	
Delone & McLean (1992)	✓		✓	✓		
Watson, Goodhue (2002)		✓			✓	✓

Fig. 1. Usability Models and Attributes

B. Global Context of Usability Evaluation

Several research projects have been carried out globally to evaluate the usability of websites in various ways. Many of them concerned hospital management systems, e-commerce websites, and websites for libraries.

In 2013, Walia and Gupta conducted a usability evaluation on the home pages of National Library websites in Asia. This study found that the National Library of Japan, which received top among the other national libraries in Asia scored 23 points out of 30 usability points [6]. Furthermore, Joo, Lin, and Lu (2011) proposed an approach for evaluating the usability of academic library websites. 18 evaluation criteria were found to measure the three characteristics of usability effectiveness, effectiveness, and learnability of academic library websites through the literature study and expert consultation [6].

Rex Hartson proposed a questionnaire-based approach to evaluating the usability of e-learning systems in 2009. The approach was created in accordance with a well-established methodology in HCI research and was based on a conceptual framework that combined Web and instructional design characteristics and linked them to the most important aspect of effective learning, which is an inherent motivation to learn [7].

A Thai university conducted a usability study of the online learning platform Microsoft Teams in 2020. The System Usability Scale (SUS), which is a Human-Computer Interaction (HCI) based technique, and the Technology Acceptance Model (TAM), which is an Information Systems (IS) based approach, were both used for evaluation purposes [8].

C. Sri Lankan Context of Usability Evaluation

There are a few published studies that evaluate library websites. One of them has been conducted to explore the factors that affected the usability of the library website of the University of Colombo. This study measured the effectiveness, efficiency, and satisfaction of the library

website through usability testing techniques and a post-test questionnaire from the user perspective [9].

In addition to the foregoing study, Rukshan & Baravalle (2011) carried out research and presented a quantitative method for evaluating the usability of websites with a focus on automated XHTML code analysis. They have created a tool by utilizing several software models, that support web usability evaluation as well and they aimed to develop an effective solution for automatically evaluating websites against web usability guidelines [10].

In 2020 the University of Sri Jayewardenepura conducted research using its LMS. The main purpose of this study was to examine the factors that affect on usability of LMS by undergraduates in higher education. This research found the effect on utility using the following five criteria, namely quality of working life, expected success, expected effort, social effects, and conditions [11].

D. Gap Identification and Variable Selection

In evaluating the usability of the MOODLE platform using ISO 9241-11 factors, a mixed-method approach incorporating both qualitative and quantitative methods was employed, using think-aloud protocol and questionnaires [12]. Unlike previous studies that used different variables to assess satisfaction, this research uniquely focused on 12 predefined tasks to measure effectiveness and efficiency. Users were invited to complete these tasks and subsequently rate their satisfaction with the platform. A significant gap identified in the existing literature is the lack of attention to users with disabilities in usability evaluations. This research addresses this gap by emphasizing the experiences and needs of users with disabilities, ensuring a more inclusive assessment of the MOODLE platform's usability.

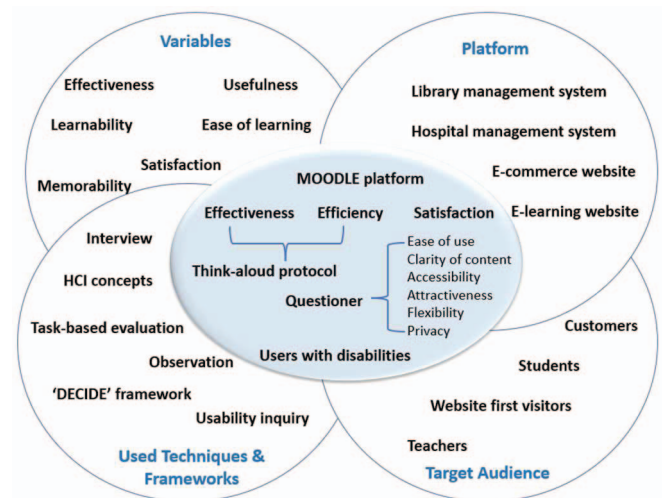


Fig. 2. Gap Identification

TABLE 1: REASON FOR VARIABLE SELECTION

Variable	Reason for selection
Ease of use	Ensure that the users can navigate and operate the MOODLE platform effortlessly, reducing the learning curve and increasing satisfaction. Nielsen, 1994; Davis, 1989

Clarity of content	Improves learning outcomes by providing clear, understandable instructions, reducing confusion, and enhancing effectiveness. Loiacono et al., 2002; DeLone & McLean, 2003
Accessibility	Guarantees the platform is inclusive and usable by all users, including those with disabilities, in line with web standards. W3C, WCAG
Attractiveness	Enhances user engagement and experience through aesthetically pleasing designs, contributing to sustained usage. Tractinsky et al., 2000
Flexibility	Supports user preferences by allowing various interaction methods, enhancing adaptability and usability. Nielsen, 1994; Dix et al., 1998
Privacy and Security	Build trust by safeguarding data and ensuring secure access, which increases user satisfaction and loyalty. Johnston et al., 2018; Li et al., 2019

III. METHODOLOGY

Usability is crucial for e-learning success, necessitating regular evaluations following proper methodologies. This study employed a mixed methods approach, integrating heuristic evaluations based on expert opinions, subjective evaluations from user opinions, and empirical evaluations from user actions. The study primarily employed a quantitative approach through a self-administered questionnaire. Additionally, the think-aloud protocol was utilized to ensure the reliability of results. The research was conducted in two stages. In the first stage, the focus was on formulating usability criteria to evaluate the usability of e-learning websites. This involved conducting a thorough literature review to study existing knowledge on usability. Additionally, expert evaluations were performed to identify new criteria and usability tasks aimed at measuring effectiveness and efficiency. Following this, in the second stage, the e-learning websites were evaluated using the formulated criteria.

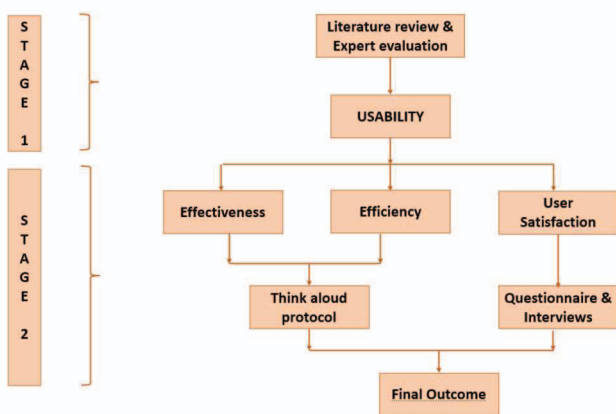


Fig. 3. Research Flow

A. Conceptual Framework

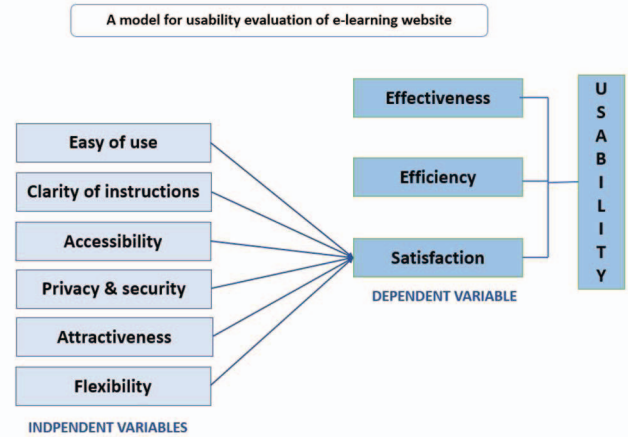


Fig. 4. Conceptual Framework

In this study, usability was measured using three dependent variables, with satisfaction being further divided into six main independent variables, which were then divided into 30 sub-variables for detailed analysis. Effectiveness and efficiency evaluation were performed via 12 tasks employing the think-aloud protocol.

B. Population & Sample Size

The study selected top-ranked state and private universities based on Webometrics rankings, focusing on MOODLE platform users with various disabilities [12]. Although the population was approximately 1000, the sample size was limited to 100 due to constraints imposed by the SmartPLS-4 student version. According to the "10-times rule," which suggests a minimum sample size of 10 times the number of arrowhead points among the dependent and independent variables, the calculated minimum sample size for this research would be 60 [13]. Therefore, the selected sample size exceeds this minimum requirement.

C. Data Collection

The data collection process began with the utilization of two separate Google forms. The first form was utilized for expert evaluation. A panel of experts was invited to select the most relevant tasks and criteria applicable for evaluating the website. Three software developers and three lecturers participated in the evaluation process, with two specializing in usability and accessibility domains.

The Google form comprised three sections. The first section solicited demographic information through an open-ended questionnaire. The second section aimed to identify criteria for evaluating user satisfaction, employing Likert scale questionnaires. The third section focused on identifying tasks and their minimum required time for measuring effectiveness and efficiency, also utilizing Likert scale questions.

Tasks for assessing effectiveness and efficiency were designed based on past studies, focusing on the primary functions typically performed on the website. Twelve tasks were categorized into two sections based on their minimum required time. Data was collected using the think-aloud protocol.

For evaluating satisfaction, ten variables were presented to the experts to select the most appropriate criteria for assessing

the website. From these, six variables were chosen for the study: Easiness, Clarity of content, Accessibility, Privacy and security, Attractiveness, and Flexibility.

D. Data Analysis

To measure the effectiveness and efficiency, descriptive analysis techniques such as the think-aloud protocol and user surveys were employed [14]. Through this protocol, website effectiveness and efficiency were measured, capturing both objective data and the subjective perspectives of respondents.

Bivariate analysis was utilized to measure user satisfaction and examine relationships among variables. This statistical analysis method is straightforward and employed to explore the relationship between two sets of variables [15].

Data collected to measure satisfaction through the questionnaire in stage 2 were exported as a CSV file. For the analysis, the study utilized Smart PLS 4, a tool for model assessment[16]. Quantitative data were analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM)[17].

PLS-SEM entails two phases of analysis: measurement model specification and structural model assessment. In the measurement model specification phase, constructs with good indicator loading, convergent validity, composite reliability, and discriminant validity are selected for further analysis. The structural model assessment phase involves weighing path coefficients and testing their significance [18].

IV. RESULTS AND DISCUSSION

The results signify a crucial phase in comprehending Moodle's usability. Centered on effectiveness, efficiency, and user satisfaction, the analysis delves deeply into user interactions. Through varied data collection methods, this section unveils pivotal findings, offering actionable recommendations for platform enhancement.

A. Analysis of Level of Effectiveness

To evaluate effectiveness and efficiency, participants were tasked with completing a set of 12 predefined tasks. These tasks were divided into two categories based on the time required for completion: those to be completed within 1 minute and those within 3 minutes. Tasks exceeding these time limits were considered incomplete.

	Task No	Task	No of respondent successfully completed the task	Percentage %	Mean value of the spent time. (Minutes)
Category 1	1	Type the Course web / LMS URL into the address bar of your browser and access the platform.	100	100	1
	2	Login to the course web/ LMS	47	47	1
	3	Navigate to the course module page	100	100	1
	4	Download the Lecture / lab sheet/ Tutorials.	50	50	1
	5	Upload the completed lab sheet/ assignment/tutorial	28	28	1
Category 2	6	To find the semester timetable	59	59	2.27
	7	To find the exam timetable & hall arrangement	57	57	2.36
	8	To find the exam results and GPA	55	55	2.32
	9	To access the announcements and events calendar.	80	80	2.06
	10	To download pass papers.	66	66	2.07
	11	To access the e-resources.	74	74	2.52
	12	To access the help desk and raise a ticket.	49	49	2.36

Fig. 5. Effectiveness & Efficiency Calculation

Based on the ISO definition of usability, effectiveness measures the accuracy and completeness of users achieving specific goals [19]. Tasks 1 and 3 were completed most

effectively with a flawless 100% success rate, while tasks 4, 6, 7, 8, 10, and 11 showed over 50% success. However, tasks 2, 5, and 12 had completion rates below 50%, indicating areas for improvement. Task 5 had the lowest success rate, impacting Moodle's overall effectiveness. The comprehensive assessment yielded an effectiveness rating of 63.75%, encompassing both task point of view and respondent point of view, revealing strengths and areas for improvement.

B. Analysis of Level of Efficiency

As per ISO standards, efficiency measures the accuracy and completeness of goals relative to resource usage [19]. In the assessment of the MOODLE platform, the focus was on task completion time, excluding unfinished tasks. The overall efficiency of the MOODLE website for the first category was 1 minute, while for the second category, it was 2.28 minutes. Similarly, each user's overall task completion time was calculated, yielding the same results.

C. Analysis of Level of Satisfaction

This chapter examines the data analysis results through PLS-SEM analysis, evaluating both the measurement and structural models. The measurement model ensures reliability and validity, while the structural model assesses the significance of the hypothesized relationships.

D. Measurement Model Analysis

Reliability, essential for stability and consistency in measurement factors, was assessed using Cronbach's Alpha and Composite Reliability methods. With Cronbach's alpha values ranging from 0.936 to 0.957 and composite reliability statistics from 0.936 to 0.967, both surpassing the 0.70 threshold, construct reliability was firmly established, ensuring robustness and consistency in the study's measurement factors [18].

The measurement model illustrates how response items relate to their underlying latent constructs. Convergent validity assesses how well multiple items measuring the same concept align. It is evaluated using three metrics: factor loadings, average variance extracted (AVE), and composite reliability (CR). Recommended thresholds are factor loadings greater than 0.5, AVE above 0.5, and CR exceeding 0.7. The results of this measurement model exceeded the recommended values, indicating sufficient convergent validity [18].

	Cronbach's alpha	Composite reliability (rho_c)	AVE
Accessibility	0.931	0.948	0.786
Attractiveness	0.957	0.967	0.852
Clarity of instruction	0.942	0.954	0.777
Ease of use	0.933	0.947	0.751
Flexibility	0.911	0.938	0.791
Privacy & security	0.954	0.966	0.878
Satisfaction	0.908	0.936	0.784

Fig. 6. Reliability and Convergent Validity Analysis

After confirming convergent validity, discriminant validity is assessed. Discriminant validity measures how well items distinguish between different constructs or concepts. It

is evaluated by comparing the square root of the average variance extracted (AVE), shown in bold on the diagonal, with the correlations between constructs. Discriminant validity is established when the bold diagonal values are higher than the off-diagonal values in their corresponding rows and columns. The study confirms discriminant validity is established by adhering to the Fornell-Larcker criterion [18] [19].

	Accessibility	Attractiveness	Clarity of instruction	Ease of use	Flexibility	Privacy & security	Satisfaction
Accessibility	0.886						
Attractiveness	0.372	0.923					
Clarity of instruction	0.655	0.671	0.882				
Ease of use	0.448	0.682	0.826	0.867			
Flexibility	0.508	0.723	0.722	0.680	0.889		
Privacy & security	0.545	0.719	0.646	0.575	0.819	0.937	
Satisfaction	0.682	0.712	0.730	0.659	0.787	0.800	0.886

Fig. 7. Discriminant Validity Analysis

1) Structural Model Analysis

The structural model ascertains the significance of hypothesized relationships. Different hypotheses were proposed to evaluate the relationship of predictors on the outcome.

H1: There is a relationship between the ease of use of MOODLE and user satisfaction.

H2: There is a relationship between the clarity of instructions on MOODLE and user satisfaction.

H3: There is a relationship between the accessibility of MOODLE and user satisfaction.

H4: There is a relationship between the attractiveness of MOODLE and user satisfaction.

H5: There is a relationship between the flexibility of MOODLE and user satisfaction.

H6: There is a relationship between the privacy and security features of MOODLE and user satisfaction.

	Original sample	Sample mean	STDEV	T STDEV	P value
Accessibility -> Satisfaction	0.326	0.320	0.087	3.752	0.000
Attractiveness -> Satisfaction	0.188	0.186	0.093	2.036	0.042
Clarity of instruction -> Satisfaction	0.309	0.295	0.088	3.504	0.000
Ease of use -> Satisfaction	0.173	0.179	0.078	2.233	0.026
Flexibility -> Satisfaction	0.208	0.210	0.103	2.017	0.044
Privacy & Security -> Satisfaction	0.270	0.270	0.129	2.084	0.037

Fig. 8. Structural Model Analysis

Interpreting the results of the hypothesis testing, the table provides an overview of the relationship between different factors (Accessibility, Attractiveness, Clarity of Instruction, Ease of Use, Flexibility, and Privacy) and user satisfaction with MOODLE. Each row represents a specific factor's impact on satisfaction, with corresponding values indicating the original sample correlation, sample mean, standard deviation (STDEV), t-standard deviation (T-STDEV), and p-value. The statistically significant p-values (all less than 0.05) indicate strong evidence to reject the null hypothesis for each factor. This suggests that there is a significant relationship between each factor and user satisfaction with MOODLE. Specifically, the correlation coefficients for Accessibility, Attractiveness, Clarity of Instruction, Ease of Use, Flexibility, and Privacy are all positive, indicating that higher levels of each factor are associated with higher levels of user

satisfaction with MOODLE. The t-standard deviations are all large relative to their standard deviations, further indicating the strength of these relationships.

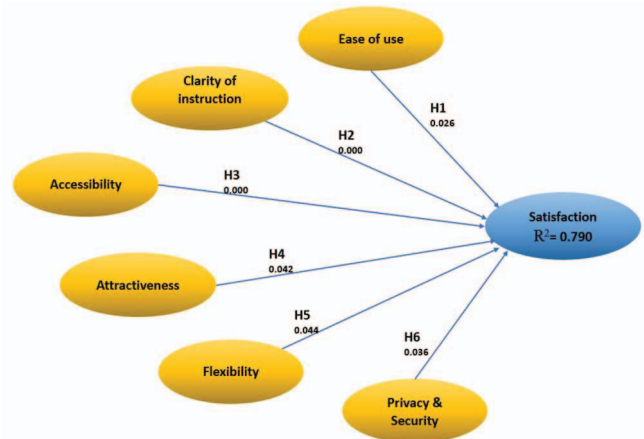


Fig. 9. Coefficient Determination

With An R2 value of 0.790, the coefficient of determination indicates that approximately 79.0% of the variability observed in "Satisfaction with the e-learning website" can be explained by the selected independent variables. This suggests a strong relationship between the independent and dependent variables, demonstrating the effectiveness of the model in predicting and understanding user satisfaction with the e-learning platform.

E. Recommendations

Based on the results and gathered data, it became evident that the existing website needs significant changes to address design, color contrast, and content issues, ensuring Moodle's usability for all users. Recommendations were developed to correct these issues, prioritized by necessity, timeframe, and resources.

Several modifications are recommended to enhance the usability of the MOODLE platform in accordance with Web Content Accessibility Guidelines (WCAG). Immediate changes should focus on establishing a well-defined heading structure and landmarks, adding appropriate navigation buttons, and addressing color contrast issues to ensure accessibility. Re-wording link names, underlining them, and providing meaningful alt text for images are crucial for better navigation and comprehension. Additionally, managing the screen length to avoid horizontal scrolling and implementing notifications for user actions will reduce confusion.

For long-term improvements, high-priority actions include introducing an early reminder feature that offers users a clear overview of upcoming activities, such as assignment deadlines. This feature can send reminders at various intervals, ensuring users are well-prepared. Another significant enhancement is adding a chatbot feature to the help desk, streamlining the process of resolving issues. Users can quickly sort out problems without needing to raise a ticket, make a phone call, or send an email.

Adding an accessibility menu to the website helps users with various disabilities navigate and use the site more easily. Features like text resizing, contrast adjustment, keyboard navigation, screen reader support, and captions/transcripts for media ensure that the website is user-friendly for everyone.

To accommodate a diverse user base, it is recommended to add a tri-lingual facility to the website. This feature will support multiple languages, making the platform more accessible to a broader audience. Additionally, other low-priority but beneficial changes include adding stop and pause buttons for animations, rewording non-meaningful links and buttons, adjusting text size, and providing a site map to aid first-time users.

Furthermore, for long-term enhancements, integrating the Learning Management System (LMS) with other university websites can significantly improve usability. Introducing a chat feature for direct communication with academic staff is also recommended. This alternative to email will facilitate more effective and timely communication, allowing students to engage not only with chatbots but also with relevant individuals, enhancing the overall communication process on the platform.

By addressing these recommendations and redesigning the MOODLE educational platform according to the WCAG, it can be made accessible not only to ordinary users but also to individuals with various disabilities, including visual, hearing, motor, and cognitive impairments. Enhancing the platform's ease of use, clarity of content, and overall attractiveness will significantly improve its usability, making MOODLE a more valuable and inclusive tool for all users.

TABLE 2: RECOMMENDED SOLUTIONS

User category	Recommended solution
Vision impairments	Provide Screen reader capabilities. Establish a well-defined heading structure and landmarks. Provide alt text for images. Add meaningful names for buttons and links. Change background and foreground colors as appropriate. Change text size as appropriate.
Hearing impairments	Provide script for video content.
Motor disabilities	Provide keyboard accessibility for all content.
Cognitive impairment	Use a simple layout. Introduce chatbot feature and site map.

V. CONCLUSION

The analysis of Moodle's usability provides a comprehensive understanding of its effectiveness, efficiency, and user satisfaction. The findings reveal both strengths and areas needing improvement, particularly emphasizing the importance of accessibility, content clarity, and ease of use to enhance the overall user experience. By implementing the actionable recommendations derived from this study, institutions can strive to create a more engaging and supportive online learning environment, ultimately enriching the educational journey for all stakeholders, including users with disabilities.

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